

Challenge (Habanero)

- Q1.** A hiker is walking at a rate of $3 \cdot \frac{\text{miles}}{\text{hour}}$ and has 2 miles left to reach the campsite. If y is the distance walked and x is the time in hours, which equation represents this situation in slope-intercept form?
- (A) $y = 3x + 2$
 (B) $y = 3x - 2$
 (C) $y = 2x + 3$
 (D) $y = 2x - 3$
 (E) $y = -3x + 2$
- Q2.** The population of deer in a forest is increasing at a rate of 2 deer per year. If the current population is 50 deer, which equation represents this situation in point-slope form?
- (A) $y - 50 = 2(x - 1)$
 (B) $y - 50 = 2(x + 1)$
 (C) $y - 50 = -2(x - 1)$
 (D) $y - 50 = -2(x + 1)$
 (E) $y - 50 = x - 2$
- Q3.** A car's fuel consumption is given by the equation $y = -\frac{1}{4}x + 12$, where y is the amount of fuel in gallons and x is the number of miles driven. Which of the following is the equation in standard form?
- (A) $\frac{1}{4}x + y = 12$
 (B) $\frac{1}{4}x - y = -12$
 (C) $-\frac{1}{4}x + y = 12$
 (D) $x + 4y = 48$
 (E) $x - 4y = -48$
- Q4.** A river's water level is dropping at a rate of 0.5 feet per hour. If the current water level is 10 feet, which equation represents this situation in slope-intercept form?
- (A) $y = -0.5x + 10$
 (B) $y = 0.5x + 10$
 (C) $y = -0.5x - 10$
 (D) $y = 0.5x - 10$
 (E) $y = -10x + 0.5$
- Q5.** A bike path has a slope of $-\frac{1}{10}$. If the path is 20 feet above the ground at $x = 0$, which equation represents this situation in point-slope form?
- (A) $y - 20 = -\frac{1}{10}(x - 1)$
 (B) $y - 20 = -\frac{1}{10}(x + 1)$
 (C) $y - 20 = -\frac{1}{10}(x - 0)$
 (D) $y - 20 = \frac{1}{10}(x - 0)$
 (E) $y - 20 = -\frac{1}{5}(x - 0)$
- Q6.** A group of friends are planning a road trip. The cost of fuel for the trip is given by the equation $y = 0.25x + 10$, where y is the cost of fuel in dollars and x is the number of miles driven. Which of the following is the equation in standard form?
- (A) $0.25x - y = -10$
 (B) $0.25x + y = 10$
 (C) $x + 4y = 40$
 (D) $x - 4y = -40$
 (E) $x + 4y = 40$
- Q7.** The altitude of a plane is given by the equation $y = -2x + 1000$, where y is the altitude in feet and x is the distance traveled in miles. If the plane is currently at an altitude of 800 feet, how many miles has it traveled?
- (A) 50 miles
 (B) 100 miles
 (C) 150 miles
 (D) 200 miles
 (E) 250 miles
- Q8.** A wind turbine's power output is given by the equation $y = 0.5x - 2$, where y is the power output in kilowatts and x is the wind speed in miles per hour. Which of the following is the equation in point-slope form?
- (A) $y - 2 = 0.5(x - 1)$
 (B) $y - 2 = 0.5(x + 1)$
 (C) $y - 2 = -0.5(x - 1)$
 (D) $y + 2 = 0.5(x - 1)$
 (E) $y + 2 = 0.5(x + 1)$
- Q9.** A boat is traveling at a rate of 5 miles per hour. If the boat has 15 miles left to reach the dock, write an equation in slope-intercept form to represent the situation. Then, convert the equation to standard form.
- Q10.** A company's profit is given by the equation $y = 2x - 1000$, where y is the profit in dollars and x is the number of units sold. Graph the equation and interpret the slope and intercepts in context. Then, convert the equation to point-slope form.

Open Ended Questions (FRQ)

- Q9.** A boat is traveling at a rate of 5 miles per hour. If the boat has 15 miles left to reach the dock, write an equation in slope-intercept form to represent the situation. Then, convert the equation to standard form.
- Q10.** A company's profit is given by the equation $y = 2x - 1000$, where y is the profit in dollars and x is the number of units sold. Graph the equation and interpret the slope and intercepts in context. Then, convert the equation to point-slope form.

Chef's Tips

- Tip 1: Provide clear instructions and examples for each question type.
- Tip 2: Allow students to use calculators for calculation-based questions.
- Tip 3: Emphasize the importance of showing work and explaining reasoning for free-response questions.